

Sumit Kumar Khadanga

Cybersecurity Engineer · Digital Forensics · Malware Analysis

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Professional Summary

Aspiring cybersecurity engineer specialising in digital forensics, malware analysis, and secure software development. Built **WinVolAuto**—a memory forensics desktop application powered by Volatility 3 with AI-driven risk scoring and MITRE ATT&CK mapping—and **CTF Decoder**, a self-learning decoding suite with a Bayesian adaptive engine deployed to Vercel. Skilled in full-stack development, ethical hacking, and blockchain-based credential systems. Seeking internships to apply offensive and defensive security expertise in real-world environments.

Technical Skills

Languages	Python, JavaScript, Java, C, SQL, HTML/CSS
Cybersecurity	Memory Forensics (Volatility 3), Malware Analysis, MITRE ATT&CK, Rootkit Detection, Ethical Hacking, OWASP Top 10, CTF Analysis
Tools	Wireshark, Nmap, Autopsy, Sleuth Kit, Burp Suite, Docker, Git, Kali Linux
Frameworks	FastAPI, Flask, React, Vue 3, PyQt6, Scikit-learn, SHAP, Streamlit
Concepts	Adaptive Heuristic Engines, Bayesian Inference, Dynamic Plugin Architecture, AI Explainability, Immutable Ledgers, Secure APIs
OS	Linux (Ubuntu, Kali), Windows

Projects

WinVolAuto – Memory Forensics Suite *Python · PyQt6 · Volatility 3 · AI Heuristics*

- Designed and developed a professional GUI for Volatility 3 to automate memory forensics across Windows, Linux, and macOS memory dumps.
- Implemented dynamic plugin discovery, context-aware configuration, and asynchronous execution via QThread to prevent UI freezing during deep scans.
- Integrated AI-powered risk analyser that scores suspicious processes using heuristics (code injection, hidden PIDs, network anomalies) and maps findings to MITRE ATT&CK techniques.
- Built triage plugins for rapid detection of persistence mechanisms, `/tmp` execution, syscall hooks, and text/file extraction from RAM.
- Generated automated PDF/JSON reports with safe table rendering and real-time plugin search for efficient forensic investigation workflows.

CTF Decoder – Adaptive Decoding Suite *Python · FastAPI · Vue 3 · SQLite · Bayesian ML · Vercel*

- Engineered a self-learning, modular decoding engine for CTF competitions supporting 30+ codecs (Base64/32/85, XOR, ROT-N, Gzip, Bzip2, Morse, URL encoding, and chained multi-layer payloads).
- Implemented a Beta-Binomial Bayesian adaptive engine that updates per-codec success priors after every solve, continuously improving detection accuracy across sessions stored in SQLite with JSONL audit logging.
- Built a multi-stage detection pipeline combining Shannon entropy analysis, character-set constraints, and English-frequency scoring (ASCII ratio, bigrams, word-match) to rank decoded candidates with confidence thresholds.
- Developed a Code Script Solver capable of static extraction and sandboxed dynamic execution of Python, C, and Java CTF scripts, piping output back through the decoder engine for automatic flag detection.
- Deployed as a Vercel serverless application with lazy ML imports, in-memory SQLite fallback for read-only FS, and a macOS Sequoia glassmorphic Vue 3 SPA; full REST API with rate-limiting via SlowAPI.

github.com/SKYLINE217/CTF-Decoder

Real-Time Credit Card Fraud Detection *Python · Scikit-learn · SHAP · Flask · Streamlit*

- Built an ML pipeline with a Random Forest classifier (optimised for high recall) and SHAP explainability to detect fraudulent transactions in real time.
- Deployed dual interfaces: a REST API (`/predict` endpoint) and a Streamlit dashboard with live demo, audit logging, and metrics visualisation.

Blockchain Academic Credential System *Python · Flask · React · MongoDB · SHA-256*

- Implemented tamper-proof credential issuance and verification using cryptographic hashing and immutable ledger principles with an append-only update mechanism.
- Designed JWT-authenticated frontend with forensic integrity checks via hash-continuity verification across the credential chain.

Experience

Cybersecurity Intern | *Edureka (Remote)* Mar 2025 – Jul 2025

- Led peer team in ethical hacking labs covering vulnerability scanning, password cracking, and red/blue team simulations.
- Completed structured curriculum on offensive security techniques, network penetration testing, and defensive countermeasures.

Education

B.Tech in Cybersecurity — NIIT University, Neemrana, Rajasthan 2022 – 2026 *Relevant Coursework:* Network Security · Ethical Hacking · Digital Forensics · Cryptography · Secure Software Development · OS · DBMS

Class XII – Science (Mathematics & Python) — Kendriya Vidyalaya ASC Centre 2022

Class X — Kendriya Vidyalaya ASC Centre 2020

Certifications

Google Cloud Career Practitioner Pathway — Google Developer Student Clubs, NIIT University 2025

- Completed curriculum covering cloud fundamentals, IAM, Compute Engine, Storage, Networking, and Security on GCP; earned official Google Cloud Certificate of Completion.

Cybersecurity and Ethical Hacking — Edureka 2025

Introduction to Cybersecurity — Cisco Networking Academy 2024